

Shock-wave from Primordial Black Hole Formation

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Introduction

- Primordial black hole (PBH) formation is accompanied by the violent expulsion of surrounding material, which might lead to the formation of shock-waves and can be regarded as a sort of “**primordial supernova**”.
- These shock-waves can generate a baryon asymmetry and may leave **observable imprints** on the cosmic microwave background (CMB) and the stochastic gravitational wave background (SGWB).
- It is possible to constrain the abundance of primordial black holes through the observable imprints of these shock-waves.
- However, different authors gave different results and explanations about whether shock-waves are generated during the formation of primordial black holes.

Numerical Method

The Misner-Sharp equations describe the motion of a spherically symmetric relativistic fluid and can be used to study PBH formation. We consider the following assumptions:

- Metric ansatz: $ds^2 = -A(t, r)^2 dt^2 + B(t, r)^2 dr^2 + R(t, r)^2 d\Omega_2^2$.
- Energy-momentum tensor of the fluid:
 $T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu}$, with $u_\mu = -A\delta_\mu^0$.
- The equation of state: $p = \omega\rho$, with $\omega = 1/3$.
- Initial curvature perturbation:

$$K(r) = \mathcal{A}\bar{K}(r) = \mathcal{A}e^{-(r/r_m)^2}, \quad (1)$$

where r_m is the length scale of the perturbation and the amplitude $\mathcal{A}$ is determined by the energy density contrast δ :

$$\mathcal{A} = \frac{\delta}{f(\omega)\bar{K}(r_m)r_m^2}, \quad f(\omega) = \frac{3(1+\omega)}{5+3\omega}. \quad (2)$$

The initial data for dynamical variables are given by the long-wavelength approximation.

With the above assumptions, the Misner-Sharp equations have the following form:

$$\begin{aligned} U &\equiv \frac{1}{A} \frac{\partial R}{\partial t}, \Gamma \equiv \frac{1}{B} \frac{\partial R}{\partial r}, M \equiv \int_0^R 4\pi R^2 \rho \left(\frac{\partial R}{\partial r} \right) dr, \\ A &= \left(\frac{\rho_b}{\rho} \right)^{\frac{\omega}{\omega+1}}, \\ \Gamma &= \sqrt{1 + U^2 - \frac{2M}{R}}, \\ \dot{U} &= -A \left[\frac{\omega}{1+\omega} \frac{\Gamma^2}{\rho} \frac{\rho'}{R'} + \frac{M}{R^2} + 4\pi R w \rho \right], \\ \dot{R} &= AU, \\ \dot{\rho} &= -A\rho(1+\omega) \left(2\frac{U}{R} + \frac{U'}{R'} \right), \\ \dot{M} &= -4\pi A w \rho U R^2. \end{aligned} \quad (3)$$

These equations can be numerically solved with the **finite difference method** in the radial direction and the **fourth-order Runge-Kutta method** in the time direction. Moreover, to avoid the breaking of the simulation due to the formation of the singularity, we excise the black hole interior region after the apparent horizon forms.

Numerical Results

The threshold for collapse in our initial data is $\delta_c \approx 0.4977$, and we studied three cases: sub-critical ($\delta = 0.45$), near-critical $\delta = 0.5$ and super-critical ($\delta = 0.55$) cases.

- Sub-critical case:** The perturbation grows and then disperses as a shock-wave.
- Near-critical case:** The perturbation grows and forms a black hole, with a shock-wave propagating outwards.
- Super-critical case:** The perturbation grows and forms a black hole, with only an underdense shell propagating outwards.

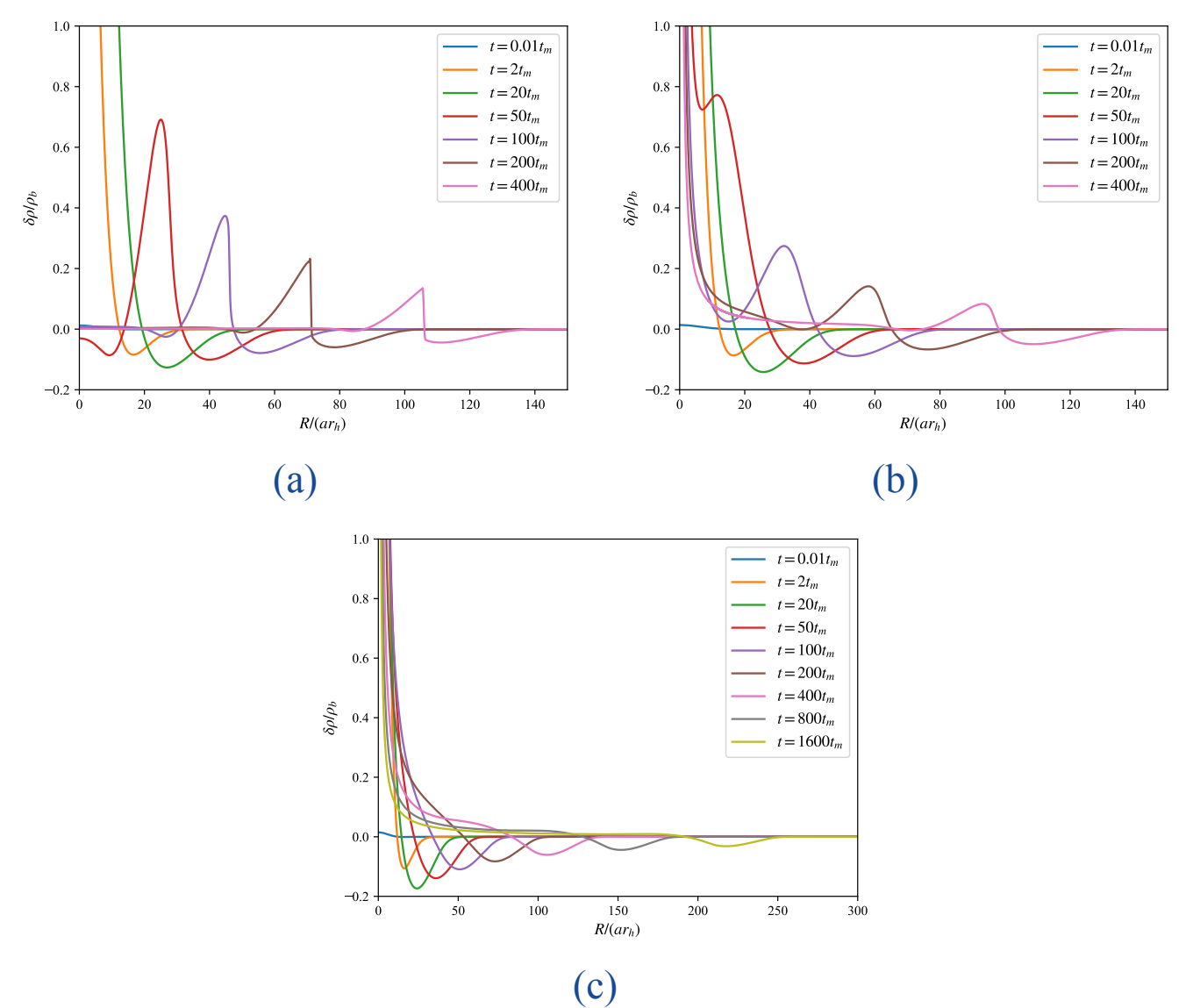


Figure 1. The time evolution of the energy density contrast for the sub-critical (a), near-critical (b), and super-critical (c) cases.

The comparison of the near-critical case with the super-critical case:

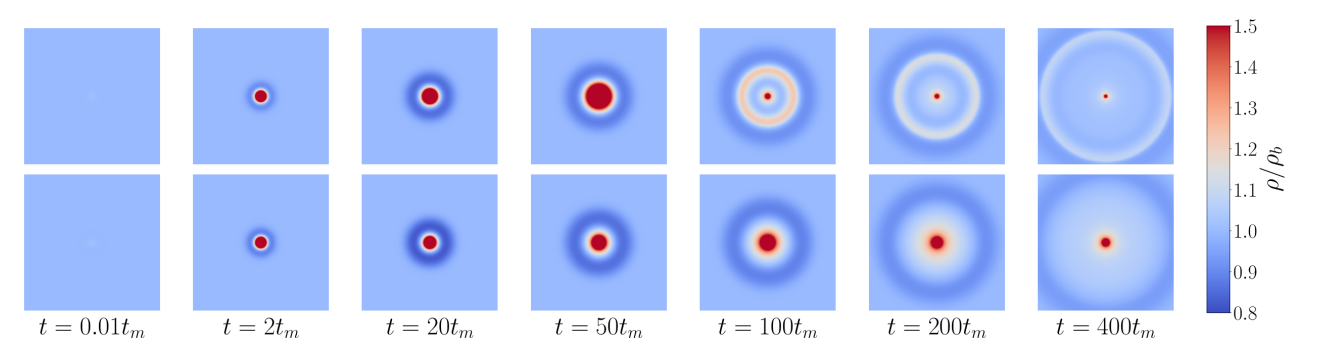


Figure 2. The comparison of the near-critical (upper panel) case with the super-critical (lower panel) case.

Considering the near-critical case, we can see that the shock-wave propagates outwards with a constant comoving width. The energy of the PBH is compensated by the overdense and underdense shells:

$$M_{\text{PBH}} + M_{\text{overdense}} + M_{\text{underdense}} = 0, \quad (4)$$

which have been checked numerically.

Conclusion

- The shock-wave formation depends on the initial perturbations, the more near-critical the perturbation is, the stronger the shock-wave forms.
- The comoving width of the shock-wave does not change with time.
- Due to the energy conservation, the overdense and underdense shells compensate for the energy of the PBH.